

Statistical Analysis Plan (SAP)

Title	Relative Effectiveness of Social Media, Dating Apps, and Information Search Sites in Promoting HIV Self-testing: Observational Cohort Study
CRU/Department/Division/Center	CTN
IRB Number	IRB# 18-001580,
Investigators:	
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Version Date	Jan 25, 2026
Project Folder Location	https://datashare.nida.nih.gov/study/nida-ctn-0083
Project Goal(s)	<i>Manuscript and possibly presentations for dissemination of comparative effectiveness results of online platforms in promoting HIV self-testing</i>

Investigator Agreement	☒ All statistical analyses included in an abstract or manuscript should reflect the work of the biostatistician(s) listed on this SAP. No changes or additional analyses should be made to the results or findings without discussing with the project biostatistician(s). ☒ All biostatisticians on this SAP should be given sufficient time to review the full presentation, abstract, manuscript, or grant and be included as co-authors on any abstract or manuscript resulting from the analyses. ☒ If substantial additional analysis is necessary or the aims of the project change, a new SAP will need to be developed. ☒ Publications resulting from this SAP are supported in part by the Duke CTSA and must cite grant number UL1TR002553 and be submitted to PubMed Central.
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☒ I have reviewed the SAP and understand that any changes must be documented.

Activity Log

1. Adaptations Due to COVID-19 Pandemic

We conducted a third wave of promotion and enrollment on Twitter, Yahoo, and Hornet. This wave was conducted between April 6, 2020, and May 6, 2020, during the first days of the public health emergency proclamation. Despite the promotional waves being active, no participants were enrolled, and no test kits were ordered during Wave 3, which made our statistical model inestimable. As enrollment during this period does not reflect “expected conditions” and scientific comparisons would not be accurate, we decided to exclude Wave 3 from all the analyses.

Acronyms

MSM	Men who have sex with men
PrEP	Pre-Exposure Prophylaxis
HIV	Human Immunodeficiency Virus
SAP	Statistical Analysis Plan
TAPS	Tobacco, Alcohol, Prescription medication, and other Substance

1 Study Overview

The study evaluated the comparative effectiveness of three types of web-based platforms—social media (Facebook, Instagram), dating apps (Grindr, Jack’D), and information search sites (Google, Bing)—in promoting free home HIV self-testing kits among minority MSM at elevated risk for HIV infection. Participants received promotional ads tailored to demographic risk groups, where test kit order rates served as a proxy for platform effectiveness.

1.1 Study Aims

Primary Aim: Compare the effectiveness of social media, dating apps, and information search sites in promoting HIV self-testing kit orders among young Black and Latinx MSM.

Secondary Aim: Evaluate associations between psychosocial factors (stigma, readiness to test, attitudes toward HIV testing and treatment, etc.) and HIV self-test kit ordering.

1.2 Study Hypotheses

1.2.1 Primary Hypotheses

H1: Dating apps will yield higher HIV self-test kit order rates than social media and information search sites.

1.2.2 Secondary Hypotheses

H2: Participants reporting higher psychological readiness to test or favorable attitudes toward HIV testing will have higher test kit order rates.

2 Study Population

2.1 Inclusion Criteria

- Cisgender MSM aged 18–30 years

- Self-identified as Black/African-American or Latinx
- Reported condomless anal sex or >1 male partner in the past 90 days

2.2 Exclusion Criteria

- Known HIV positive status
- Tested for HIV in the past 90 days
- Currently taking PrEP or used PrEP within past six months

2.3 Data Acquisition

Fill in all relevant information:

Study design	Longitudinal observational cohort study with web-based advertising recruitment.
Data source/how the data were collected	Online platform metrics, baseline survey responses, and follow-up self-reports via web surveys.
Contact information for team member responsible for data collection/acquisition	(no publicly available information)
Date or version (if downloaded, provide date)	Recruitment occurred through multiple waves around 2020.
Data transfer method and date	(no publicly available information)
Where dataset is stored	(no publicly available information)

Description:

Recruitment ads were run in waves across three web platforms at the same time, targeting high HIV incidence regions in the U.S. Eligible participants completed baseline surveys and could order a free HIV self-test kit for home use.

3 Outcomes, Exposures, and Additional Variables of Interest

3.1 Primary Outcome(s)

Outcome	Description	Variables and Source	Specifications
HIV Self-Test Kit Order Rate	Daily number of HIV self-test kits ordered through each platform type	1. Data field for number of orders can be calculated by adding: "ora_within60_yesno". 2. The number of days per site: first wave 70, second wave 38	1. Coding specification for orders is: Yes; No; Missing. 2. The rate can be calculated as Rate = n_orders / n_days (per site)

3.2 Secondary Outcome(s)

Outcome	Description	Variables and Source	Specifications
Psychosocial Correlates	Relationship between attitudes/stigma and likelihood of ordering. It will	1. Tobacco, alcohol, prescription medications, and other substance use 2. Stage of Health Behavior Change	Encoded by categorical/continuous type as per validated scales.

	be analyzed via 6 dimensions.	3. Attitudes toward human immunodeficiency virus (HIV) testing 4. Attitudes toward human immunodeficiency virus (HIV) treatment 5. Human immunodeficiency virus (HIV)-related stigma among study participants 6. Medical mistrust	1. Levels: None, Problem use, High Risk Substance use 2. Binary: Yes=1, No=2 3. Levels: Agree=1, Disagree=2 4. Nums 5. Levels: Strongly agree=1, Agree=2, Somewhat agree=3, Neither agree nor disagree=4, Somewhat disagree=5, Disagree=6, Strongly disagree=7 6. Levels: Strongly agree=1, Agree=2, Disagree=3, Strongly disagree=4
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3.3 Additional Variables of Interest

Variable	Description	Variables and Source	Specifications
Q14_2	I feel afraid of people living with HIV/AIDS.	Num, Baseline survey	1=Strongly agree; 2=Agree; 3=Somewhat agree; 4=Neither agree nor disagree; 5=Somewhat disagree; 6=Disagree; 7=Strongly disagree
Q16_1	You'd better be cautious when dealing with healthcare organizations.	Num, Baseline survey	28=Strongly agree; 30=Agree; 33=Disagree; 34=Strongly disagree
Q3_1	How old are you?	From Baseline survey	Numerical value

4 Statistical Analysis Plan

4.0 Overview of Analysis Approach

1. Pre-Analysis Data Preparation

- Define analytic sample: participants enrolled when all three platform types (social media, dating apps, information search sites) were active concurrently.
- Exclude participants from periods when platforms were not simultaneously active (e.g., early wave when Grindr was inactive).
- Define ordered kit vs not ordered kit based on whether participants ordered a self-test kit within 60 days of receiving an order code.

2. Outcome Definitions

- Primary outcome: Rate of HIV self-test kits ordered per day during the advertisement period for each platform type.
- Secondary outcomes: Associations between psychosocial and behavioral variables with test kit ordering.

3. Platform Grouping

- Platforms were categorized by type: 1. Social media: Facebook, Instagram, 2. Dating apps: Grindr, Jack'D, 3. Information search sites: Google, Bing

4.1 Demographic and Clinical Characteristics ("Table 1")

1. **Purpose:** Summarize participant characteristics to understand the cohort and identify potential covariates for secondary analyses.

2. **Variables to summarize:**

- Demographics: age, race, ethnicity
- Sexual behavior: number of male sex partners (past 90 days), condom use patterns
- HIV testing history: ever tested, months since last test
- Substance use risk scores: measures for tobacco, alcohol, cannabis, etc.
- Psychosocial measures: attitudes toward HIV testing/treatment, stigma items, medical mistrust scales

3. **Statistics:**

- **Continuous variables:** summarize their means and standard deviations and median with interquartile ranges to capture distribution shapes.
- **Categorical variables:** Frequencies and percentages.

4.2 Analyses Plan for Aim 1

1. **Primary Objective:** Compare HIV self-test kit order rates across platforms.
2. **Poisson Regression Modeling:** Use a Poisson regression model to compare ordering rates across platform types. The dependent variable is count of kits ordered, offset is the number of days for each promotion period.
3. **Calculation of Order Rates:** For each platform site, compute: Order rate=Number of kits ordered / Number of advertising days in wave
4. **Pairwise Comparisons:** Conduct pairwise comparisons of order rates among platforms within each wave, there is a total of 6 pair.
5. **Adjusted methods:** To address the issue that significant platform-by-wave interactions and widely differing order rates between sites within the same platform, compare the specific platform differences in terms of the order rates within the same wave, using the Hochberg method.
6. **Make Interpretation:** Evaluate whether order rates differ statistically between sites within the same wave. Use P-values to identify if there is significant rate differences.

4.3 Analyses Plan for Aim 2

1. **Objective:** Assess whether psychosocial or behavioral measures differ between participants who did and did not order a self-test kit.
2. **Measurements to examine:**
 - Psychological readiness to test
 - Attitudes toward HIV testing and treatment
 - HIV-related stigma
 - Medical mistrust
 - Substance use risk
3. **Handle Missing Data from survey**

Participants who skipped questions will be coded as “missing” and excluded from specific analyses.

4. Comparisons:

Group by people who ordered kits and did not order kits. Then calculate summary statistics: mean and std for continuous variable, count and percentage for categorical variable. Also perform statistical tests: If likert variable: use Wilcox test. If categorical: use Fisher’s exact test. If Continuous, use Student’s t test.

5 Limitations

The study’s findings should be interpreted with caution for several reasons. First, data were collected from only nine areas with high HIV incidence, which limits how well the results can be generalized to the broader population. Second, low enrollment and participation across waves reduced the ability to make strong comparisons between platforms or across different sites. Finally, the study focused on a set of popular apps and websites grouped into platform categories, meaning the results apply specifically to the sites included in the campaigns rather than all possible digital platforms.

6 References

1. Stafylis C, et al. Relative Effectiveness of Social Media, Dating Apps, and Information Search Sites in Promoting HIV Self-testing: Observational Cohort Study. JMIR Formative Research. 2022;6(9):e35648.
2. CTN Research Brief on the study (CTN-0083).